Room-temperature quantum sensing for magnetic imaging: A Multidisciplinary Research Program

Specific Aims

Aim 1: Advance room-temperature quantum sensing for magnetic imaging via a spectroscopy-focused strategy.

Aim 2: Advance room-temperature quantum sensing for magnetic imaging via a spectroscopy-focused strategy.

Background

The approach integrates established spectroscopy methods with a novel analytical pipeline. Prior work has established a partial understanding, yet key gaps remain. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement.

Approach

Findings will be disseminated through peer-reviewed publication and open data release. Prior work has established a partial understanding, yet key gaps remain. The approach integrates established spectroscopy methods with a novel analytical pipeline. This proposal addresses room-temperature quantum sensing for magnetic imaging, a problem with substantial significance.

Innovation

Our preliminary data suggest a tractable path toward measurable improvement.

Investigator Context

The applicant is a community nonprofit partnering with an academic core. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Budget: personnel $426,114; equipment $90,807; travel $13,213; indirect costs $134,662. Total requested support is $664,796 over 3 years at a R1 institution.

References

1. [1] https://doi.org/10.1021/jacs.9b02765
2. [2] https://doi.org/10.1073/pnas.1517384113
3. [3] https://doi.org/10.1016/j.cell.2016.07.054
4. [4] https://doi.org/10.1126/science.1259855
5. [5] https://doi.org/10.1109/TPAMI.2016.2577031